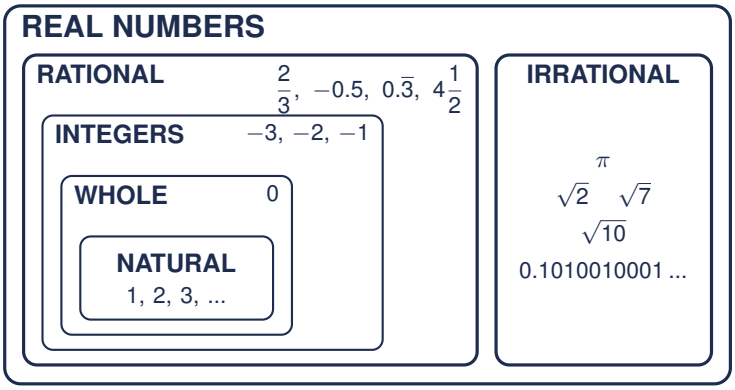
The Real Number System

NC.8.NS.1 • NC.8.NS.2 • NC.8.EE.2 (square roots)

Unit 1 • Front

1 Real Numbers: How the Sets Fit Together



Natural	Counting numbers: 1, 2, 3, ...
Whole	Natural numbers **and** 0
Integers	Whole numbers and their opposites: ..., $-2, -1, 0, 1, 2, ...$
Rational	Can be written as a fraction $\frac{a}{b}$ (a, b integers, $b \neq 0$). Decimal **ends** or **repeats**.
Irrational	**Cannot** be a fraction. Decimal **never ends** and **never repeats**.

Every number has a decimal form.
The decimal tells you the type!

2 Rational or Irrational? Check the Decimal

Number	Decimal	Why	Type
$\frac{3}{4}$	0.75	Decimal ends (terminates)	**Rational**
$\frac{1}{3}$	$0.333... = 0.\overline{3}$	Decimal repeats	**Rational**
$\sqrt{49}$	7	49 is a perfect square (7×7)	**Rational**
$-\sqrt{16}$	-4	16 is a perfect square (4×4)	**Rational**
$\sqrt{10}$	3.16227766 ...	10 is **not** a perfect square. Never ends, never repeats.	**Irrational**
π	3.14159265 ...	Never ends, never repeats	**Irrational**

3 Fraction → Decimal: Divide!

Numerator ÷ Denominator

$\frac{\text{top}}{\text{bottom}} \rightarrow \text{bottom}\,)\overline{\text{top}}$

1. Top number goes **inside**.
2. Add a decimal point and zeros.
3. Divide. Bring the decimal straight up.
4. Stop when:
   remainder = 0 → terminates
   remainder repeats → repeats

Terminating: $\frac{3}{8}$

$$\begin{array}{r} 0.375 \\ 8\,)\overline{3.000} \\ -24 \\ \hline 60 \\ -56 \\ \hline 40 \\ -40 \\ \hline 0 \end{array}$$

Remainder 0. Stop!

$\boxed{\frac{3}{8} = 0.375}$

Repeating: $\frac{5}{11}$

$$\begin{array}{r} 0.4545 \\ 11\,)\overline{5.0000} \\ -44 \\ \hline 60 \\ -55 \\ \hline 50 \\ -44 \\ \hline 60 \\ -55 \\ \hline 5 \end{array}$$

Remainders: 6, 5, 6, 5, ... repeat!

$\boxed{\frac{5}{11} = 0.4545... = 0.\overline{45}}$

Bar notation: The bar goes over **only** the digits that repeat. $0.\overline{45} = 0.454545...$ $0.1\overline{6} = 0.16666...$ $2.\overline{3} = 2.333...$

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4 Repeating Decimal → Fraction: The 9s Method

1. **Top:** write the digits that repeat.
2. **Bottom:** write one **9** for each repeating digit.
3. **Simplify.** Divide top and bottom by the GCF.
4. **Whole number?** Keep it in front (mixed number).

Challenge: repeat starts late. $0.1\overline{6}$
Top: (digits up to one repeat) − (digits that don't repeat) = 16 − 1 = 15
Bottom: a **9** for each repeating digit, then a **0** for each non-repeating digit = 90

$$0.1\overline{6} = \frac{15}{90} = \frac{15 \div 15}{90 \div 15} = \boxed{\frac{1}{6}}$$

Decimal	Use the 9s	Simplify
$0.\overline{7}$	$\frac{7}{9}$	$\boxed{\frac{7}{9}}$ (already simplest)
$0.\overline{36}$	$\frac{36}{99}$	$\frac{36 \div 9}{99 \div 9} = \boxed{\frac{4}{11}}$
$0.\overline{123}$	$\frac{123}{999}$	$\frac{123 \div 3}{999 \div 3} = \boxed{\frac{41}{333}}$
$2.\overline{3}$	$2\frac{3}{9}$	$2\frac{3 \div 3}{9 \div 3} = \boxed{2\frac{1}{3}}$

5 Perfect Squares and Square Roots



Square: multiply a number by **itself**.
$7^2 = 7 \times 7 = 49$

Perfect square: a whole number squared.
1, 4, 9, 16, 25, 36, ...

Square root $\sqrt{\ }$**:** the number that was squared.
$\sqrt{49} = 7$

They undo each other!
$7 \xrightarrow{\text{square}} 49 \xrightarrow{\sqrt{\ }} 7$
Square = **area**. Square root = **side**.

Solve $x^2 = p$: $x = \pm\sqrt{p}$ (2 answers!)

$x^2 = 81$ $\quad x = \pm 9$
$x^2 = \frac{4}{9}$ $\quad x = \pm\frac{2}{3}$
$x^2 = 7$ $\quad x = \pm\sqrt{7}$ (irrational)

Why ±? $9 \times 9 = 81$ and $(-9) \times (-9) = 81$

1^2	2^2	3^2	4^2	5^2	6^2	7^2	8^2	9^2	10^2	11^2	12^2	13^2	14^2	15^2	16^2	17^2	18^2	19^2	20^2
1	4	9	16	25	36	49	64	81	100	121	144	169	196	225	256	289	324	361	400

6 Estimate Non-Perfect Square Roots (to the Tenths)

Method 1: Number Line $\sqrt{50}$

1. Trap it between perfect squares.
   $49 < 50 < 64$
2. Take the square roots.
   $\sqrt{49} = 7$ and $\sqrt{64} = 8$
3. Is 50 closer to 49 or 64?
   Much closer to 49 → just past 7
4. Place it on the number line.

$\boxed{\sqrt{50} \approx 7.1}$

Pi: $\pi \approx 3.14$ (to the hundredths)

Least to greatest:
$\sqrt{3} < \sqrt{8} < \pi < \sqrt{50}$

Method 2: The 4-Step Algorithm $\sqrt{8}$

$$\sqrt{n} \approx \text{lower root} + \frac{n - \text{lower square}}{\text{upper square} - \text{lower square}}$$

1. **Trap it:** $4 < 8 < 9$
   roots: 2 and 3
2. **Fraction:** $\frac{8-4}{9-4} = \frac{4}{5}$
3. **Divide:** $4 \div 5 = 0.8$
4. **Add:** $2 + 0.8 = 2.8$

$\boxed{\sqrt{8} \approx 2.8}$

3 Divide

$$5\overline{)\,4.0}\quad 0.8,\quad -40,\quad 0$$

Check: square it

$$\begin{array}{r} 2.8 \\ \times\ 2.8 \\ \hline 224 \\ +\ 560 \\ \hline 7.84 \end{array}$$

$7.84 \approx 8$ ✓

